

ETC2410 Assignment1

Group 88

Question 1

(a)

The estimated ordinary least squares (OLS) regression equation is:

$$\widehat{\text{math10}} = 13.390 + 0.002 \cdot \text{expend} \quad (1)$$

(2.936) (0.001)

(b)

For a \$100 increase:

$$\Delta \text{math10} = 0.002 \times 100 = 0.2$$

The estimated coefficient $\hat{\beta}_1 = 0.002$ indicates that a \$100 increase in expenditure is associated with a 0.2 percentage point increase in math10, holding other factors constant.

(c)

Yes, it does. The positive coefficient of $\hat{\beta}_1$ aligns with expectations, as higher investment in schools is generally expected to improve educational outcomes. However, the magnitude of the effect is very small, suggesting that while spending has a positive influence, other factors likely play a more significant role in determining math pass.

(d)

$$\widehat{\text{math10}} = 25.064 + 0.002 \cdot \text{expend} - 0.0002 \cdot \text{enroll} - 0.307 \cdot \text{lnchprg} \quad (2)$$

(3.077) (0.001) (0.0002) (0.035)

(e)

In the simple regression (Q1a), $\hat{\beta}_1$ measures the total association between expenditure and math10, without accounting for other factors. In the multiple regression (Q1d), $\hat{\beta}_1$ still represents the effect of expenditure, but now it is the ceteris paribus effect, holding constant enrollment and the lunch program variable.

(f)

If enrollment increases by **1,000 students**, then:

$$\Delta \text{math10} = -0.0002 \times 1000 = -0.2$$

The estimated coefficient $\hat{\beta}_1 = -0.0002$ indicates that an increase in enrollment by 1,000 students is associated with a 0.2 percentage point decrease in math10 score, holding other factors constant.

(g)

From the regression output,

Expend ($p = 0.0137 < 0.05$)

lnchprg ($p < 2.2\text{e-}16 < 0.05$)

Enroll ($p = 0.4177 > 0.05$)

Based on the regression results, expend and lnchprg are individually significant at the 5% significance level ($p < 0.05$), while enroll is not significant ($p > 0.05$). This means that, after controlling for the other regressors, only the significant variables (expend and lnchprg) have a statistically reliable effect on math10, while enroll does not show strong evidence of an effect.

(h)

At the school level, additional factors that may help explain *math10* include measures of teacher quality (such as certification, experience, and student–teacher ratios), student demographics (socioeconomic background, language proficiency, or minority status), and levels of parental involvement. Broader school resources and community socioeconomic conditions may also contribute to variation in student achievement.

Question 2

(a)

Null hypothesis $H_0: \beta_{\text{Lngchprg}} = 0$

Alternative hypothesis $H_1: \beta_{\text{Lngchprg}} < 0$

Test Statistic and Sampling Distribution:

$$t = \frac{\hat{\beta}_{\text{lnchprg}} - 0}{SE(\hat{\beta}_{\text{lnchprg}})} = -8.655$$

The t-statistic follows a t-distribution with degrees of freedom equal to: $n - k - 1 = 404$

Decision rule: Reject H_0 if $t < t_{\text{critical}}$, otherwise, fail to reject H_0

Sample and Critical Values:

From the regression output, the sample t-value is $t = -8.654592$

The critical value for a left-tailed test at the 5% significance level with 404 degrees of freedom is $t_{(0.05, 404)} = -1.648634$ (calculated using R: `qt(0.05, 404)`).

Conclusion: Since $-8.654592 < -1.648634$, we reject the null hypothesis. There is sufficient evidence at the 5% significance level to conclude that the percentage of students in the school lunch program has a statistically significant negative effect on average pass rates, after controlling for school expenditure and enrollment.

(b)

Unrestricted model:

$$\text{math10} = \beta_0 + \beta_1 \cdot \text{expend} + \beta_2 \cdot \text{enroll} + \beta_3 \cdot \text{lnchprg} + u$$

Restricted model:

$$\text{math10} = \beta_0 + \beta_1 \cdot \text{expend} + u$$

Null hypothesis $H_0: \beta_2 = 0$ and $\beta_3 = 0$

Alternative hypothesis H_1 : At least one of β_2 and β_3 is not zero

Test Statistic and Sampling Distribution:

The test statistic is the F-statistic for joint significance:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}$$

The F-statistic follows an F-distribution with degrees of freedom: $F_{q, n-k-1} = F_{(2, 404)}$

Sample and Critical Values:

From the R output, the sample F-statistic is 37.45282

The critical value is 3.018056 (calculated using `qf(0.95, 2, 404)` in R).

Decision rule: Reject H_0 if F-statistic > critical F-value, otherwise, fail to reject H_0

Conclusion: Since $37.45282 > 3.018056$, we reject the null hypothesis. There is sufficient evidence at the 5% significance that at least one of the variables—school enrollment or the percentage of students in the school lunch program—has a significant effect on math pass rates, after controlling for per-student expenditure.

(c)

Null hypothesis H_0 : $\beta_1 = \beta_2 = \beta_3 = 0$

Alternative hypothesis H_1 : At least one coefficient is not zero

Test Statistic and Sampling Distribution:

The test statistic is the F-statistic:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}$$

The F-statistic follows an F-distribution with degrees of freedom $(q, n - k - 1) = (3, 404)$

Sample and Critical Values:

From the regression output, the sample F-statistic is 30.38582

The critical value is 2.626992 (calculated using `qf(0.95, 3, 404)` in R).

Decision Rule : If $F_{sample} > F_{critical}$ reject H_0 , otherwise, fail to reject H_0

Conclusion:

Since F-statistic (30.38582) is greater than the critical value (2.626992) we reject the null hypothesis at the 5% significance level. There is sufficient evidence to conclude that at least one of the regressors (expenditure per student, school enrollment, or percentage of students in the school lunch program) has a statistically significant effect on math pass rates.

Question 3

(a)

Original Model and Hypothesis

$$math10 = \beta_0 + \beta_1 \cdot expend + \beta_2 \cdot enroll + \beta_3 \cdot lnchprg + u$$

Null hypothesis $H_0 : \beta_1 = \beta_2$

Alternative hypothesis $H_1 : \beta_1 > \beta_2$

Model Transformation

Let $\theta = \beta_1 - \beta_2$ then $\beta_2 = \beta_1 - \theta$

$$math10 = \beta_0 + \beta_1 \cdot expend + (\beta_1 - \theta) \cdot enroll + \beta_3 \cdot lnchprg + u$$

$$math10 = \beta_0 + \beta_1 \cdot (expend + enroll) - \theta \cdot enroll + \beta_3 \cdot lnchprg + u$$

Let $\gamma = -\theta$:

Transformed Model and Hypothesis

$$math10 = \beta_0 + \beta_1 \cdot (expend + enroll) + \gamma \cdot enroll + \beta_3 \cdot lnchprg + u$$

Null hypothesis $H_0 : \gamma = 0$ (expenditure has the same effect as enrollment)

Alternative hypothesis $H_1 : \gamma < 0$ (expenditure has a larger effect as enrollment)

(b)

Model

$$math10 = \beta_0 + \beta_1 \cdot (expend + enroll) + \gamma \cdot enroll + \beta_3 \cdot lnchprg + u$$

Null hypothesis $H_0 : \gamma = 0$ (expenditure has the same effect as enrollment)

Alternative hypothesis $H_1 : \gamma < 0$ (expenditure has a larger effect as enrollment)

Test Statistic and Sample distribution

$$t = \frac{\hat{\gamma} - 0}{SE(\hat{\gamma})}$$

The t-statistic follows a t-distribution with degrees of freedom equal to: $n - k - 1 = 404$

Sample and critical values:

From the regression output, the sample t-value is $t = -2.384852$

The critical value is $t_{(0.05, 404)} = -1.648634$ (calculated using R: `qt(0.05, 404)`).

Decision Rule: Reject H_0 if the $t_{statistic} < t_{critical}$, otherwise, fail to reject H_0

Conclusion: Since the sample t-statistic (-2.384852) is greater than the critical value (1.648624), we reject H_0 . There is sufficient evidence at the 5% significance level to conclude that expenditure per student has a larger effect on math pass rates than school enrollment.

(c)

Original Model

$$math10 = \beta_0 + \beta_1 \cdot expend + \beta_2 \cdot enroll + \beta_3 \cdot lnchprg + u$$

Under H_0 : $\beta_1 = \beta_2$ we can substitute it into the original model:

$$math10 = \beta_0 + \beta_1 \cdot (expend + enroll) + \beta_3 \cdot lnchprg + u$$

Unrestricted vs. Restricted Models

$$math10 = \beta_0 + \beta_1 \cdot expend + \beta_2 \cdot enroll + \beta_3 \cdot lnchprg + u$$

$$math10 = \beta_0 + \beta_1 \cdot (expend + enroll) + \beta_3 \cdot lnchprg + u$$

F-test Specification

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}$$

(d)

Null hypothesis $H_0 : \beta_1 = \beta_2$ (Expenditure per student has the same effect as school enrollment)

Alternative hypothesis $H_1 : \beta_1 \neq \beta_2$ (The effects are different)

Test statistic and its sampling distribution:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}$$

The F-statistic follows an F-distribution with degrees of freedom $(q, n - k - 1) = (1, 404)$

Sample and critical values:

From the R code output below, the sample F-statistic is 5.68752

The critical value for the F-distribution is 3.864579

Decision Rule: If $F_{sample} > F_{critical}$ reject H_0 , otherwise, fail to reject H_0

Conclusion: Since the sample F-statistic (5.68752) is greater than the critical value (3.864579), we reject H_0 . There is sufficient evidence at the 5% significance level to conclude that the effects of expenditure per student and school enrollment on math pass rates are significantly different.

Question 4

(a)

Define a dummy variable *staffhigh* where

$$\text{staffhigh} = \begin{cases} 1 & \text{if } \frac{\text{staff members}}{1000 \text{ students}} > 100 \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$\text{math10} = \beta_0 + \beta_1 \cdot \text{expend} + \beta_2 \cdot \text{enroll} + \beta_3 \cdot \text{lnchprg} + \beta_4 \cdot \text{staffhigh} + \beta_5 \cdot (\text{enroll} \times \text{staffhigh}) + u$$

(b)

The estimated Extended Model regression equation is:

$$\widehat{\text{math}} = 25.154 + \underset{(3.289)}{0.002} \cdot \text{expend} - \underset{(0.0003)}{0.0005} \cdot \text{enroll} - \underset{(0.036)}{0.301} \cdot \text{lnchprg} + \underset{(1.427)}{2.203} \cdot \text{staff high} + \underset{(0.0004)}{0.0005} \cdot (\text{enroll} \times \text{staffhigh}) + u \quad (3)$$

(c)

Unrestricted Model (U):

$$\text{math10} = \beta_0 + \beta_1 \cdot \text{expend} + \beta_2 \cdot \text{enroll} + \beta_3 \cdot \text{lnchprg} + \beta_4 \cdot \text{staffhigh} + \beta_5 \cdot (\text{enroll} \times \text{staffhigh}) + u$$

Restricted Model (R):

$$\text{math10} = \beta_0 + \beta_1 \cdot \text{expend} + \beta_2 \cdot \text{enroll} + \beta_3 \cdot \text{lnchprg} + u$$

Null hypothesis $H_0 : \beta_4 = 0$ and $\beta_5 = 0$

Alternative hypothesis H_1 : At least one of β_4 and β_5 is not zero

Decision Rule: If $F_{\text{sample}} > F_{\text{critical}}$ reject H_0 , otherwise, fail to reject H_0

Test statistic and its sampling distribution:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} \sim F_{q, n-k-1}$$

The F-statistic follows an F-distribution with degrees of freedom $(q, n - k - 1) = (2, 402)$

Sample and Critical Values:

From the R output, the sample F-statistic is 1.265594

The critical value is 3.018168

Conclusion:

Since $F = 1.265594 < 3.018168$, we fail to reject the null hypothesis. There is insufficient evidence at the 5% significance level to conclude that `staff_high` has an effect on `math10`.

(d)

Based on the modified estimated extended regression model(3):

$$\widehat{\text{math}} = 25.154 + 0.002 \cdot \text{expend} - 0.0005 \cdot \text{enroll} - 0.301 \cdot \text{lnchprg} + 2.203 \cdot \text{staffhigh} + 0.0005 \cdot (\text{enroll} \times \text{staffhigh}) + u$$

(3)

Plug in expend = 5,332, enroll = 4,046, lnchprg = 5.5

Let staffhigh=0 for 90 staff/1000 or staffhigh=1 for 120 staff/1000

From the R output :

School A prediction percentage : 32.1395

School B prediction percentage: 36.3655

Under identical conditions of per-student expenditure, enrollment size, and school lunch program participation, **School B** (with a higher student-to-staff ratio of 120 per 1000) is predicted to have a **higher math pass rate (36.37%)** compared to **School A (32.14%)** with 90 staff per 1000 students. This suggests a positive association between increased staffing levels and improved math performance, as captured by the model.

Appendix

Question 1

(a)

```
# Load required package and dataset
library(wooldridge)

load("C:/Users/Matilda Fang/Desktop/2025-S2/ETC2410 - ETC5241 - Introductory econometrics/ETC2410-ETC5241 Data")

# Estimate the OLS regression model:  $\text{math10} = \beta_0 + \beta_1 \text{expend} + u$ 
model_1 <- lm(math10 ~ expend, data=df)

# Display model summary
summary(model_1)
```

Call:

```
lm(formula = math10 ~ expend, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-22.572	-7.224	-0.848	6.410	39.160

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.339e+01	2.936e+00	4.560	6.78e-06 ***
expend	2.452e-03	6.606e-04	3.712	0.000234 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.34 on 406 degrees of freedom

Multiple R-squared: 0.03282, Adjusted R-squared: 0.03044

F-statistic: 13.78 on 1 and 406 DF, p-value: 0.0002345

(d)

```
# Load required packages
library(wooldridge)
library(modelsummary)

# Load dataset
data("df")

# Estimate multiple regression model
model_2 <- lm(math10 ~ expend + enroll + lnchprg, data = df)

# Display model summary
summary(model_2)
```

Call:

```
lm(formula = math10 ~ expend + enroll + lnchprg, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
--	-----	----	--------	----	-----

-24.536 -6.182 -1.164 4.884 41.939

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	25.0642875	3.0772101	8.145	4.77e-15 ***
expend	0.0016445	0.0006643	2.476	0.0137 *
enroll	-0.0001543	0.0001902	-0.811	0.4177
lnchprg	-0.3069202	0.0354633	-8.655	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.519 on 404 degrees of freedom

Multiple R-squared: 0.1841, Adjusted R-squared: 0.178

F-statistic: 30.39 on 3 and 404 DF, p-value: < 2.2e-16

(g)

```
library(wooldridge)
data("df")

model_2 <- lm(math10 ~ expend + enroll + lnchprg, data = df)

summary(model_2)
```

Call:

lm(formula = math10 ~ expend + enroll + lnchprg, data = df)

Residuals:

Min	1Q	Median	3Q	Max
-24.536	-6.182	-1.164	4.884	41.939

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	25.0642875	3.0772101	8.145	4.77e-15 ***
expend	0.0016445	0.0006643	2.476	0.0137 *
enroll	-0.0001543	0.0001902	-0.811	0.4177
lnchprg	-0.3069202	0.0354633	-8.655	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.519 on 404 degrees of freedom
Multiple R-squared: 0.1841, Adjusted R-squared: 0.178
F-statistic: 30.39 on 3 and 404 DF, p-value: < 2.2e-16

```
significant_vars <- names(coef(model_2))[summary(model_2)$coefficients[,4] < 0.05]  
significant_vars
```

```
[1] "(Intercept)" "expend"      "lnchprg"
```

Question 2

(a)

```
# Estimate the model  
model <- lm(math10 ~ expend + enroll + lnchprg, data = df)  
summary_model <- summary(model)  
# Get coefficient and standard error for lnchprg  
beta_lnchprg <- coef(summary_model)["lnchprg", "Estimate"]  
se_lnchprg <- coef(summary_model)["lnchprg", "Std. Error"]  
  
# Calculate t-statistic  
t_value <- beta_lnchprg / se_lnchprg  
  
# get df  
df_residual <- summary_model$df[2]  
  
# Critical value for one-sided test (alpha=0.05, left-tailed)  
critical_t <- qt(0.05, df = df_residual)  
  
# Output results  
cat("Estimated coefficient for lnchprg:", beta_lnchprg, "\n")
```

Estimated coefficient for lnchprg: -0.3069202

```
cat("Standard error for lnchprg:", se_lnchprg, "\n")
```

Standard error for lnchprg: 0.03546328

```
cat("t-statistic for lnchprg:", t_value, "\n")
```

t-statistic for lnchprg: -8.654592

```
cat("Critical t-value (alpha=0.05, df=", df_residual, "):", critical_t, "\n")
```

Critical t-value (alpha=0.05, df= 404): -1.648634

(b)

```
# Estimate unrestricted model
unrestricted_model <- lm(math10 ~ expend + enroll + lnchprg, data = df)
SSR_ur <- sum(residuals(unrestricted_model)^2)

# Estimate restricted model
restricted_model <- lm(math10 ~ expend, data = df)
SSR_r <- sum(residuals(restricted_model)^2)

# Calculate F-statistic
F_test <- anova(restricted_model, unrestricted_model)
F_stat <- F_test[2, "F"]

# Calculate critical value
critical_value <- qf(0.95, 2, 404) # df1 = 2, df2 = 404

# Display result
cat("F-statistic:", F_stat, "\n")
```

F-statistic: 37.45282

```
cat("Critical value (5% level):", critical_value, "\n")
```

Critical value (5% level): 3.018056

(c)

```

# Estimate the unrestricted model
model <- lm(math10 ~ expend + enroll + lnchprg, data = df)

# Get the summary to extract F-statistic
summary_model <- summary(model)

# Extract components for F-test
SSR <- sum(residuals(model)^2)
SST <- sum((df$math10 - mean(df$math10))^2)
SSE <- SST - SSR
q <- 3
k <- 3
n <- nrow(df)

# Calculate F-statistic manually
F_stat_manual <- (SSE / q) / (SSR / (n - k - 1))

# Alternatively, get F-statistic from summary
F_stat <- summary_model$fstatistic[1]

# Critical value
critical_value <- qf(0.95, q, n - k - 1)

# Display results
cat("SST:", SST, "\n")

```

SST: 44866.37

```
cat("SSR:", SSR, "\n")
```

SSR: 36606.56

```
cat("SSE:", SSE, "\n")
```

SSE: 8259.804

```
cat("F-statistic (manual):", F_stat_manual, "\n")
```

F-statistic (manual): 30.38582

```
cat("F-statistic (from summary):", F_stat, "\n")
```

F-statistic (from summary): 30.38582

```
cat("Critical value (5%):", critical_value, "\n")
```

Critical value (5%): 2.626992

```
cat("Decision:", ifelse(F_stat > critical_value, "Reject H0", "Fail to reject H0"), "\n")
```

Decision: Reject H0

Question 3

(b)

```
# Create the new combined variable
df$expend_plus_enroll <- df$expend + df$enroll

# Estimate the transformed model
model_transformed <- lm(math10 ~ expend_plus_enroll + enroll + lnchprg, data = df)
summary(model_transformed)

# Extract gamma coefficient and its standard error
gamma_estimate <- coef(model_transformed)["enroll"]
gamma_se <- sqrt(vcov(model_transformed)["enroll", "enroll"])

# Calculate t-statistic and p-value (one-tailed test)
t_statistic <- gamma_estimate / gamma_se
p_value_one_tailed <- pt(t_statistic, df = df$residual(model_transformed))

# Critical value for one-tailed test at 5% significance level
t_critical <- qt(0.05, df = df$residual(model_transformed), lower.tail = TRUE)

# Display results
cat("t-statistic:", t_statistic, "\n")
cat("One-tailed p-value:", p_value_one_tailed, "\n")
cat("t_critical:", t_critical)
```

(c)

```
# Unrestricted model (original)
unrestricted <- lm(math10 ~ expend + enroll + lnchprg, data = df)

# Restricted model (under H0:  $\beta_1 = \beta_2$ )
df$expend_plus_enroll <- df$expend + df$enroll
restricted <- lm(math10 ~ expend_plus_enroll + lnchprg, data = df)

# F-test
anova(restricted, unrestricted)
```

(d)

```
# Unrestricted model (original)
unrestricted <- lm(math10 ~ expend + enroll + lnchprg, data = df)

# Restricted model (under H0:  $\beta_1 = \beta_2$ )
df$expend_plus_enroll <- df$expend + df$enroll
restricted <- lm(math10 ~ expend_plus_enroll + lnchprg, data = df)

# Perform F-test
f_test_result <- anova(restricted, unrestricted)

# Extract results
ssr_r <- sum(residuals(restricted)^2)
ssr_u <- sum(residuals(unrestricted)^2)
f_statistic <- f_test_result$F[2]
p_value <- f_test_result$`Pr(>F)`[2]
critical_value <- qf(0.95, 1, 404) # = 0.05

# Display results
cat("F-test Results:\n")
```

F-test Results:

```
cat("Restricted Model SSR:", round(ssr_r, 2), "\n")
```

Restricted Model SSR: 37121.91


```
cat("Unrestricted Model SSR:", round(ssr_u, 2), "\n")
```

Unrestricted Model SSR: 36606.56

```
cat("F-statistic:", f_statistic, "\n")
```

F-statistic: 5.68752

```
cat("Critical value (=0.05):", critical_value, "\n")
```

Critical value (=0.05): 3.864579

Question 4

(b)

```
# Create staff_high dummy variable
df$staff_high <- ifelse(df$staff > 100, 1, 0)

# Estimate the extended model
extended_model <- lm(math10 ~ expend + enroll + lnchprg + staff_high + enroll:staff_high, data = df)

# Get model summary
model_summary <- summary(extended_model)
coefficients <- coef(model_summary)
summary(extended_model)
```

Call:

```
lm(formula = math10 ~ expend + enroll + lnchprg + staff_high +
    enroll:staff_high, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-23.698	-6.101	-0.843	4.880	42.204

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept)      25.1539236  3.2895523   7.647 1.54e-13 ***
expend           0.0018917  0.0007306   2.589 0.00997 **
enroll          -0.0004914  0.0003017  -1.629 0.10413
lnchprg         -0.3006910  0.0361185  -8.325 1.33e-15 ***
staff_high      -2.2029477  1.4274133  -1.543 0.12354
enroll:staff_high 0.0004853  0.0003646   1.331 0.18393
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.513 on 402 degrees of freedom
Multiple R-squared:  0.1892,    Adjusted R-squared:  0.1791
F-statistic: 18.76 on 5 and 402 DF,  p-value: < 2.2e-16

```

c

```

# Unrestricted model (from 4b)
unrestricted <- lm(math10 ~ expend + enroll + lnchprg + staff_high + enroll:staff_high, data = df)

# Restricted model (excluding staff variables)
restricted <- lm(math10 ~ expend + enroll + lnchprg, data = df)

# F-test
f_test <- anova(restricted, unrestricted)

# Extract results
f_statistic <- f_test$F[2]
p_value <- f_test$`Pr(>F)`[2]
critical_value <- qf(0.95,2,402) # = 0.05

# Display results
cat("F-statistic:", f_statistic, "\n")

```

F-statistic: 1.265594

```
cat("p-value:", p_value, "\n")
```

p-value: 0.2831932

```
cat("Critical value:", critical_value, "\n")
```

Critical value: 3.018168

(d)

```
# math = 25.154 + 0.002*expend - 0.0005*enroll - 0.301*lnchprg
#          + 2.203*staff_high + 0.0005*(enroll*staff_high)

expend <- 5332
enroll <- 4046
lnchprg <- 5.5

predict_math <- function(expend, enroll, lnchprg, staff_high){
  25.154 +
  0.002 * expend -
  0.0005 * enroll -
  0.301 * lnchprg +
  2.203 * staff_high +
  0.0005 * (enroll * staff_high)
}

# schoolA: 90 staff/1000 (staff_high = 0)
schoolA <- predict_math(expend, enroll, lnchprg, staff_high = 0)

# schoolB: 120 staff/1000 (staff_high = 1)
schoolB <- predict_math(expend, enroll, lnchprg, staff_high = 1)

cat("School A prediction:", schoolA, "\n")
```

School A prediction: 32.1395

```
cat("School B prediction:", schoolB, "\n")
```

School B prediction: 36.3655